

R Practice 6 – Graphing with Base R

This assignment is just for your own practice and does not need to be submitted. If you have any questions, please feel free to reach out to me, come to office hours, or schedule another a time to meet with me.

Open an R Script in RStudio and complete each of the following:

1. **College Distance.** On Canvas is the data set **CollegeDistance.csv** that was collected in the 1980s on high school students. This data set contains the following variables:
 - gender:** factor indicating gender.
 - ethnicity:** factor indicating ethnicity (African-American, Hispanic or other).
 - score:** base year composite test score. These are achievement tests given to high school seniors in the sample.
 - fcollege:** factor. Is the father a college graduate?
 - mcollege:** factor. Is the mother a college graduate?
 - home:** factor. Does the family own their home?
 - urban:** factor. Is the school in an urban area?
 - unemp:** county unemployment rate in 1980.
 - wage:** state hourly wage in manufacturing in 1980.
 - distance:** distance from 4-year college (in 10 miles).
 - tuition:** average state 4-year college tuition (in 1000 USD).
 - education:** number of years of education.
 - income:** factor. Is the family income above USD 25,000 per year?
 - region:** factor indicating region (West or other).
- a. Read this data set into **R**.
- b. Create a bar plot of the **gender** variable. To do this, you will need to use the **table()** function inside of the **barplot()** function. Give it an x and y label as well as a title. Feel free to make the bars look nicer as well!
- c. Create a pie chart of the **ethnicity** variable (just to see how bad it is). Again, the **table()** function will be needed here. Adjust the labels (with the **labels** option in the function) so that they are “African American”, “Hispanic”, and “Other”. Then change the colors of the pie chart so the African American slice is blue, the Hispanic slice is green, and the Other slice is orange.
- d. Create a scatter plot with **tuition** on the y-axis and **unemp** on the x-axis. Change the x and y labels to be “Unemployment Rate” and “Average 4-Year College Tuition (in thousands)”. Change the plot so that the points are filled-in dark blue dots. Then change the size of the labels to be 1.3 and put a title.

- e. Create a histogram of the **distance** variable. Change the x-axis label and the title of the plot so it looks nicer. Also, change the color from gray to light blue.
- f. Create side-by-side boxplots for the **score** variable split by **income**. Change the x-axis and y-axis labels and give the plot a title. Change the color of the boxplots to one of your choice.

2. **Nile River Flow.** The file **nile.csv** is on Canvas and contains information about the annual flow of the Nile River at Aswan measured in 10^8 cubic meters from 1871 to 1970.

- a. Read the data set into **R**.
- b. Consider the plot below. Try to recreate it as closely as you can.

